# From Static to Contextual

## Distributional Semantics and What's Inside BERT

Two independent implementations, one set of conclusions

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Common presentation

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## The problem

| Task 1 – baseline | Task 2 – reproduction | Task 3 – combine |
|---|---|---|
| PPMI + SVD vectors from EWT train: one fixed vector per word. | Hewitt & Manning structural probe on frozen `bert-base-uncased`, layers 1–12 $\Rightarrow$ best syntax layer $\ell^*$. | Average layer-$\ell^*$ vectors over all contexts $\Rightarrow$ static vectors; compare with Task 1. |

### Evaluated on

**meaning**: WordSim-353 Spearman, Google analogies

**syntax**: MST over word distances vs. the gold tree (UUAS)

### Two questions

(1) Does pooling from the syntax layer give better vectors – and do meaning and syntax agree?
(2) Because we implemented everything twice: **which conclusions survive a change of implementation?**

# Two implementations: what is shared, what differs

## Shared

- EWT splits: 12,544 / 2,001 / 2,077 sentences
- WS-353 + Google analogies (gensim copies)
- frozen `bert-base-uncased`, word = mean of pieces
- paper probes and L1 losses, best-dev checkpoint
- $\ell^*$ chosen on dev

| | **A** (Y. D. Babu) | **B** (T. Saha) |
|---|---|---|
| T1 vocabulary | all 16,654 types | 8,865 types (count $\geq$2) |
| WS "all 353" | OOV pair $\rightarrow$ median sim. | OOV pair $\rightarrow$ 0 |
| 20-sentence MST | in-vocab, punctuation excluded | any words, punctuation included |
| larger MST set | 660 eligible sentences | 1,926 test sentences |
| **probe lr** | $\times$**0.1 on dev plateau (paper)** | **constant** $10^{-3}$ |
| T3 vocabulary | all types | T1's 8,865 types (+ all types) |
| anisotropy fix | all-but-the-top ($D$=7) | mean-centring, + PCA-100 |
| uncertainty | bootstrap CIs, paired tests | micro/macro, full test split |

# Code organisation of the two implementations

## A: 23CS10081_Yellapu_Dheeraj_Babu/code/

| | |
|---|---|
| `common/` | data, MST, stats, shared eval |
| `task1/build_ppmi_svd.py` | co-occurrence → PPMI → SVD |
| `task2/extract_bert.py` | 13-layer feature cache |
| `task2/train_probes.py` | 78 probes, lr decay |
| `task3/pool_static.py` | mean per word type |
| `task3/evaluate_task3.py` | same eval + paired tests |
| `run_all.py` + Colab | one command |

## B: Codes/ + results/

| | |
|---|---|
| `common.py` | EWT `.conllu`, benchmarks, MST |
| `Task1.py` | co-occurrence → PPMI → `svds` |
| `Task2.py` | features, `probe_loss`, `train_probe` (layers trained in chunks) |
| `Task3.py` | static vectors, centring/PCA, sweep |
| `results/` | JSON/CSV outputs, figures |

## Same repository, clearly separated

`github.com/trishitasaha/NLP_Assignment2`: **B**'s work at the root (`Codes/`, `results/`), **A**'s work in its own folder, the common report and slides in `Combined/`.

# Cross-check: same settings ⇒ same numbers

| quantity | **A** in **B**'s setting | **B** reported |
|---|---|---|
| types with count $\geq 2$ | 8,865 | 8,865 |
| analogy questions, all words count $\geq 2$ | 3,406 | 3,406 |
| T3 WS-353 $\rho$ on those 203 pairs | **0.648** | **0.648** |
| T3 WS-353 $\rho$, all 353, missing $= 0$ | **0.288** | **0.288** |
| T3 WS-353 $\rho$, 248 covered pairs | 0.577 | 0.577 |
| T3 analogy, 5,727 covered / all | 67.7 / 19.8% | 67.7 / 19.8% |
| T1 WS-353 $\rho$ on the 203 pairs | 0.171 | 0.189 |

## Reading

- The two Task 3 pipelines are **equivalent**: identical to three decimals.
- Task 1 differs only because **B** drops words seen once before the SVD.
- Remaining differences are *settings*, not bugs.

# Task 1: both find weak, data-limited vectors

| Task 1 PPMI-SVD | A | B |
|---|---|---|
| WS-353 $\rho$, covered pairs | 0.104 (248) | 0.189 (203) |
| WS-353 $\rho$, all 353 | 0.048 (median) | 0.024 (zero) |
| analogy acc., covered | 0.51% | 0.85% |
| MST UUAS, 20 sentences | 20.5% | 16.7% |
| random / chain | 15.2 / 43.2 | 13.1 / 35.5 |
| MST UUAS, larger set | 24.8% | 17.8% |

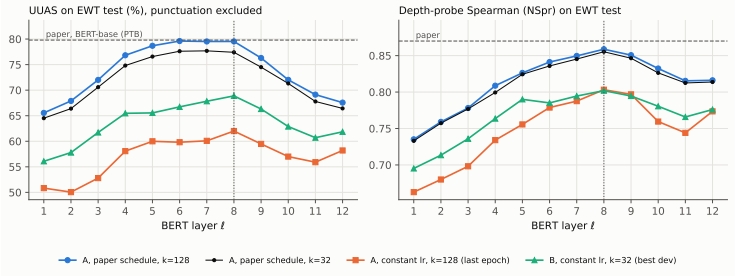
## Agreement

- $\rho$ near 0 on all 353 pairs; analogies $<1\%$
- MST: above random, below the linear chain
- cause = 200k words (**A**: $\rho$ 0.317 on frequent pairs)

## Why B's covered $\rho$ is higher

Its 203 pairs exclude rare words: **A**'s vectors score 0.171 on them.

# Task 2: same best layer, different absolute UUAS




$\ell^* = 8$

in both implementations

| | UUAS | NSpr |
|---|---|---|
| paper (PTB) | 79.8 | 0.87 |
| **A**, lr decay | 79.5 | 0.859 |
| **B**, constant lr | 68.9 | 0.802 |
| **A**, constant lr | 62.0 | 0.803 |

UUAS with punctuation excluded.

## Why the UUAS differs: the learning-rate schedule

### Evidence

- **A**'s control with **B**'s schedule (constant lr) drops to 62.0% (last epoch); **B** keeps the best dev epoch: 68.9%.
- **B**: best dev epoch of $k$=128 probes is mostly epoch 1. **A**'s control: dev minimum at epoch 1, then it oscillates for 29 epochs.
- **A**'s control: *training* loss also stays high (0.62 vs. 0.42 with decay) $\Rightarrow$ **non-convergence, not over-fitting**.

### Mechanism

L1 loss: the gradient does not shrink near the optimum, so Adam keeps taking steps of size $\approx$ lr and circles the minimum. The paper's $\times 0.1$ decay lets it settle.

### It also flips the best rank

**B** (constant lr): $k$=32 best. **A** (decay): $k$=128 $\geq$ 64 $>$ 32 (77.4 / 79.2 / 79.5%), as in the paper.

# Task 3, meaning: both find a large gain

| | | WS $\rho$ (203 pairs) | analogy, covered |
|---|---|---|---|
| **A** | T1 PPMI-SVD | 0.171 | 0.51% |
| **B** | T1 PPMI-SVD | 0.189 | 0.85% |
| **A** | T3 BERT-static, layer 8 | **0.648** | **67.7%** |
| **B** | T3 BERT-static, layer 8 | **0.648** | **77.1%** |
| **A** | T3 + all-but-the-top | 0.716 | 71.0% |
| **B** | T3 mean-centred | 0.706 | 78.3% |

**B**'s analogy questions and candidates use its 8,865-word vocabulary, which raises covered accuracy.

## Agreement

- WS $\rho$ $\approx$0.18 $\rightarrow$ 0.65; analogies $<$1% $\rightarrow$ 68–77%
- **A**'s paired tests: $\Delta\rho = +0.474$ [0.36, 0.61]; 3,848 vs. 0 questions
- removing the shared direction helps in both (anisotropy)
- not data-matched: BERT was pre-trained on $\approx$3.3B words

# Task 3, trees: both find no improvement

| UUAS (%) | 20 sentences A | 20 sentences B | larger set A (660) | larger set B (1,926) |
|---|---|---|---|---|
| random tree | 15.2 | 13.1 | 17.1 | 14.3 |
| linear chain | 43.2 | 35.5 | 43.0 | 40.5 |
| T1 PPMI-SVD | 20.5 | 16.7 | 24.8 | 17.8 |
| T3 BERT-static | **18.4** | **9.6** | **17.9** | **14.5** |
| T3 + anisotropy fix | 19.7 | 13.5 | 19.6 | 16.0 |
| T2 probe (A) | 81.2 | – | 82.9 | – |

**Agreement**

T3 < T1 in both implementations, on both sets – close to a random tree.

**Significance (A)**

20 sentences: $\Delta = -2.1$ pt, *n.s.*
660 sentences: $\Delta = -6.9$ pt $[-8.0, -5.8]$

**So**

**meaning and syntax disagree** – in both.

# Why meaning and syntax disagree: B's reasons, A's measurements

| B: explanation | A: measurement |
|---|---|
| Pooling removes the context-dependent signal | ladder on 660 sentences: static 17.9% → through probe $B$ 35.9% → contextual cosine 38.2% → probe **82.9%** |
| The probe learns a projection; the MST uses raw cosine | even contextual vectors with raw cosine (38.2%) stay below the linear chain (43.0%) |
| Edges join semantically dissimilar words; PPMI captures co-occurrence | T3 neighbours share the POS 0.74 (T1 0.33) = substitutes; T1 neighbours co-occur 0.56 (T3 0.16) = partners |
| | does closer mean connected? edge AUC: T3 **0.46** · T1 0.54 · contextual 0.70 · probe **0.96** |
| OOV words inject arbitrary distances (B's sample: 7.4% of words) | A's in-vocabulary sample avoids this, and T3 is still below T1 |

## In one sentence

Benchmarks reward similarity between word *types* (substitutes); trees need relations between *tokens* in context (partners) – exactly what pooling removes.

## What survives a change of implementation?

### Robust: both agree

- $\ell^* = 8$, mid-layer peak, decline to layer 12
- T3 $\gg$ T1 on WordSim-353 and analogies
- T3 $<$ T1 on the MST check, near random
- static semantics peaks at layers 1–2, not $\ell^*$
- centring / all-but-the-top helps T3
- T3 numbers identical once settings match

### Implementation-sensitive

- probe UUAS: 62–80 (learning-rate schedule)
- best probe rank: 32 vs. 128 (same cause)
- T1 $\rho$: 0.10 vs. 0.17–0.19 (pair subset, cut-off)
- MST levels: up to 8 pt (sentence sample, punctuation)
- "all 353" $\rho$: imputation rule

Lesson: report these choices explicitly – some move numbers more than the effects being studied.

## Conclusions

1. **T1:** classical vectors from 200k words are weak and data-limited (both).
2. **T2:** BERT encodes dependency trees in a low-rank subspace of its middle layers; $\ell^* = 8$ (both). With the paper's schedule the probe reproduces the paper: 79.5% vs. 79.8%.
3. **T3:** pooling the syntax layer gives **much better word-meaning vectors** but **worse tree structure** (both).
4. **The two evaluations disagree** because they measure different things: type similarity vs. token relations in context.
5. **Two implementations** turned a single result into a robust one, and exposed the choices that matter (lr schedule, evaluation subsets).

**A good static meaning vector is not automatically a good syntactic representation.**

**Thank you** Questions?

# References


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- M.-C. de Marneffe et al. (2021). Universal Dependencies. Computational Linguistics 47(2).


Full details: common report (`Combined/report_combined.pdf`); individual reports in each author's folder.